Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$s = r\theta$$

Trig

$$\sin\theta = \frac{y}{r}, \quad \cos\theta = \frac{x}{r}$$

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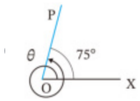
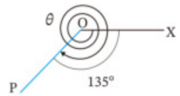
Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q001 **Written**

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus $75°$; (b) two full clockwise revolutions and a further $135°$.



(b)



Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q002 **Written**

Draw the terminal side for $300°$ and $-450°$.

Q003 **Written**

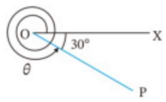
Express $420°$ **and** $-210°$ **as** $\alpha + 360° \times n$, **where** n **is an integer and** $0° \leq \alpha < 360°$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

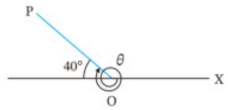
Q004 **Written**

Find the measures of θ in the two figures: (a) a clockwise turn of $30°$ from the positive x-axis; (b) a terminal side $40°$ above the negative x-axis, measured counter-clockwise from the positive x-axis).

a)



(b)



2 Draw the terminal side for the following angles.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q005 **Written**

Draw the terminal side for $500°$ and $-120°$.

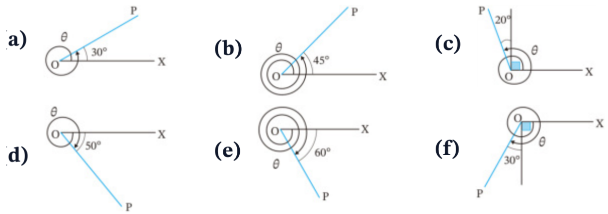
Q006 **Written**

Express $520°$ **and** $-140°$ **as** $\alpha + 360° \times n$**.**

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q007 **Written**

Find the six diagram measures θ shown in the given figure (a)–(f).



2 Draw the terminal side for the following angles.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q008 Written

Draw the terminal sides for $470°, 780°, 225°, -380°, -990°, -200°$.

Q009 **Written**

Express $610°, 920°, 840°, -430°, -960°, -20°$ **as** $\alpha + 360° \times n$**.**

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q010 **Written**

Express $129720°$ **and** $-2592100°$ **as** $\alpha + 360° \times n$, **with** $0° \leq \alpha < 360°$.

Q011 *MCQ*

The principal angle coterminal with $765°$ is:

Choose the correct option.

A $45°$

B $135°$

C $225°$

D $315°$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q012 MCQ

The principal angle coterminal with $-410°$ is:

Choose the correct option.

A $-50°$

B $50°$

C $310°$

D $410°$

Q013 **MCQ**

The terminal side of $540°$ lies on the:

Choose the correct option.

A positive x-axis

B negative x-axis

C positive y-axis

D negative y-axis

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q014 MCQ

A positive angle of $315°$ terminates in:

Choose the correct option.

A QI

B QII

C QIII

D QIV

Q015 *MCQ*

Which is the general-angle form for $-75°$?

Choose the correct option.

A $75° + 360°$ n

B $285° + 360°$ n

C $-285° + 360°$ n

D $360°\, n - 75°$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q016 MCQ

The direction of a negative angle is:

Choose the correct option.

A counter-clockwise

B clockwise

C radial only

D undefined

Q017 **MCQ**

The terminal side of $1080°$ is the:

Choose the correct option.

A positive x-axis

B negative x-axis

C positive y-axis

D negative y-axis

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q018 MCQ

For $\alpha = 0°$, the general-angle family is:

Choose the correct option.

A $360°n$

B $0°n$

C $180° + 360°n$

D $90° + 360°n$

Q019 *MCQ*

The principal angle for $-990°$ is:

Choose the correct option.

A $90°$

B $180°$

C $270°$

D $-270°$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q020 MCQ

A ray (P) is $25°$ above the negative x-axis. Its standard-position angle is:

Choose the correct option.

A $25°$
B $155°$
C $205°$
D $335°$

Q021 **Written**

(1) Convert $630°$ to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q022 **Written**

Complete the standard-angle degree/radian table, then: convert $288°, -400°$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q023 **Written**

Convert $108^\circ, 36^\circ, -240^\circ, -225^\circ, 420^\circ, 400^\circ, -570^\circ, -2700^\circ$ **to radians.**

Q024 **Written**

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ **to degrees.**

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q025 **Written**

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Q026 **Written**

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

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Q027 Written

A sector of radius $9\,\mathrm{cm}$ and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Q028 MCQ

Convert $150°$ to radians:

Choose the correct option.

A $\frac{5\pi}{6}$

B $\frac{3\pi}{5}$

C $\frac{5\pi}{3}$

D $\frac{\pi}{6}$

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Q029 **MCQ**

Convert $-\frac{5\pi}{6}$ to degrees:

Choose the correct option.

A $-120°$

B $-150°$

C $150°$

D $-300°$

Q030 MCQ

For $r = 5, \theta = \frac{2\pi}{3}$, the arc length is:

Choose the correct option.

A $\frac{5\pi}{3}$

B $\frac{10\pi}{3}$

C $\frac{25\pi}{3}$

D 10π

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q031 MCQ

For $r = 4, \theta = \frac{\pi}{2}$, the sector area is:

Choose the correct option.

A π

B 2π

C 4π

D 8π

Q032 MCQ

The radian measure of a full turn is:

Choose the correct option.

A π

B 2π

C 180

D 360

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q033 *MCQ*

The degree measure of $\frac{7\pi}{4}$ is:

Choose the correct option.

A 210°

B 270°

C 315°

D 360°

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Choose the correct option.

A $\frac{\pi}{6}$

B $-\frac{\pi}{6}$

C $\frac{5\pi}{6}$

D $\frac{11\pi}{6}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q035 MCQ

For $r = 3, \theta = \pi$, the sector area is:

Choose the correct option.

A $\frac{3\pi}{2}$
B 3π
C 9π
D $\frac{9\pi}{2}$

Q036 MCQ

The angle $\frac{5\pi}{3}$ lies in:

Choose the correct option.

A QI
B QII
C QIII
D QIV

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q037 **Written**

Convert $252°$ to radians.

Q038 **Written**

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S.

Classified Practice Defining Angles and Basic Trigonometric Functions

Q039 Written

If $\theta = \frac{4\pi}{3}$, **find** $\sin\theta, \cos\theta, \tan\theta$.

Q040 **Written**

Find the exact values (write undefined where appropriate): $\sin\frac{7\pi}{4}, \cos\frac{\pi}{3}, \tan\frac{\pi}{6}, \sin(-\frac{2\pi}{3}), \cos(-\pi), \tan(-\frac{\pi}{2})$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q041 **Written**

Find the exact values of the eighteen trigonometric functions listed in the given exercise (a)–(r), including negative and undefined cases.

Find the values of the following trigonometric functions. If the value is undefined, write “undefined”.

(a) $\sin\frac{\pi}{3}$ **(b)** $\sin\frac{7}{6}\pi$ **(c)** $\sin\pi$

(d) $\cos\frac{5}{4}\pi$ **(e)** $\cos\frac{5}{3}\pi$ **(f)** $\cos\frac{7}{6}\pi$

(g) $\tan\frac{2}{3}\pi$ **(h)** $\tan\frac{5}{4}\pi$ **(i)** $\tan\pi$

(j) $\sin\left(-\frac{\pi}{6}\right)$ **(k)** $\sin\left(-\frac{5}{4}\pi\right)$ **(l)** $\sin\left(-\frac{5}{3}\pi\right)$

(m) $\cos\left(-\frac{\pi}{4}\right)$ **(n)** $\cos\left(-\frac{5}{6}\pi\right)$ **(o)** $\cos\left(-\frac{4}{3}\pi\right)$

(p) $\tan\left(-\frac{4}{3}\pi\right)$ **(q)** $\tan\left(-\frac{11}{6}\pi\right)$ **(r)** $\tan\left(-\frac{3}{2}\pi\right)$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q042 **Written**

The panel height is $h(t) = 3\sin(\frac{2\pi}{3}t - \frac{\pi}{4}) + 1.5$. Find the exact height at $t = 0$.

Q043 MCQ

$\sin\frac{\pi}{6}$ **equals:**

Choose the correct option.

A 0

B $\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D 1

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q044 MCQ

$\cos\frac{2\pi}{3}$ **equals:**

Choose the correct option.

A -1
B $-\frac{1}{2}$
C $\frac{1}{2}$
D $\frac{\sqrt{3}}{2}$

Q045 MCQ

$\tan \frac{3\pi}{4}$ **equals:**

Choose the correct option.

A -1

B 0

C 1

D undefined

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q046 **MCQ**

$\sin(-\frac{\pi}{2})$ **equals:**

Choose the correct option.

A -1

B 0

C 1

D undefined

Q047 **MCQ**

$\tan\frac{\pi}{2}$ **is:**

Choose the correct option.

A 0

B 1

C -1

D undefined

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q048 MCQ

$\cos\frac{5\pi}{3}$ **equals:**

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Q049 MCQ

If $P(x, y)$ lies on the unit circle, then $\sin\theta$ is:

Choose the correct option.

A x

B y

C x/y

D 1/x

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q050 MCQ

$\tan(-\frac{3\pi}{4})$ **equals:**

Choose the correct option.

A -1

B 0

C 1

D undefined

Q051 MCQ

$\sin\frac{11\pi}{6}$ **equals:**

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q052 **Written**

Find $\sin \frac{5\pi}{6}, \cos \frac{4\pi}{3}, \tan \frac{7\pi}{4}$.

Q053 **Written**

Find $\sin(-\frac{5\pi}{6}), \cos(-\frac{7\pi}{6}), \tan(-\frac{\pi}{3})$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q054 **Written**

For $P = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$ on the unit circle, find the corresponding $\sin\theta, \cos\theta, \tan\theta$.

Q055 **Written**

Find the exact value of $3\sin\frac{3\pi}{2} - 2\cos\pi$**.**

Classified Practice Defining Angles and Basic Trigonometric Functions

Q056 Written

State the two standard angles where $\tan\theta$ is undefined in $[0, 2\pi)$.

Q057 **Written**

Solve for θ, $0 \leq \theta < 2\pi$: (a) $\sin\theta = -\frac{1}{2}$; (b) $\cos\theta = \frac{1}{\sqrt{2}}$; (c) $\tan\theta = -\sqrt{3}$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q058 **Written**

Solve the six trigonometric equations in the given Try set for $0 \le \theta < 2\pi$: $\sin\theta = 1/\sqrt{2}$, $\cos\theta = -\sqrt{3}/2$, $\tan\theta = 1/\sqrt{3}$, $\sin\theta = 0$, $\cos\theta = 1/2$, $\tan\theta = -1$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q059 Written

Solve the sixteen equations in the given exercise set (a)–(p) for $0 \le \theta < 2\pi$.

Q060 **Written**

Solve $\tan^2\theta + (1-\sqrt{3})\tan\theta = \sqrt{3}$ **for** $0 \le \theta < 360°$**.**

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q061 MCQ

For $0 \leq \theta < 2\pi$, $\sin\theta = \frac{1}{2}$ gives:

Choose the correct option.

A $\frac{\pi}{6}, \frac{5\pi}{6}$

B $\frac{\pi}{3}, \frac{2\pi}{3}$

C $\frac{\pi}{6}, \frac{7\pi}{6}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Q062 MCQ

For $0 \leq \theta < 2\pi$, $\cos\theta = -\frac{1}{2}$ gives:

Choose the correct option.

A $\frac{\pi}{3}, \frac{5\pi}{3}$

B $\frac{2\pi}{3}, \frac{4\pi}{3}$

C $\frac{\pi}{6}, \frac{5\pi}{6}$

D $\frac{4\pi}{3}, \frac{5\pi}{3}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q063 MCQ

For $0 \leq \theta < 2\pi$, $\tan\theta = 1$ gives:

Choose the correct option.

A $\frac{\pi}{4}, \frac{5\pi}{4}$

B $\frac{3\pi}{4}, \frac{7\pi}{4}$

C $\frac{\pi}{4}, \frac{7\pi}{4}$

D $\frac{3\pi}{4}, \frac{5\pi}{4}$

Q064 **MCQ**

The solutions of $\sin\theta = 0$ **in** $[0, 2\pi)$ **are:**

Choose the correct option.

A $0, \pi$
B $\frac{\pi}{2}, \frac{3\pi}{2}$
C $0, 2\pi$
D π

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q065 MCQ

If $\cos\theta = 1$, then in $[0, 2\pi)$, θ is:

Choose the correct option.

A 0
B $\frac{\pi}{2}$
C π
D $\frac{3\pi}{2}$

Q066 MCQ

For $0 \leq \theta < 2\pi$, $\tan\theta = -\sqrt{3}$ **gives:**

Choose the correct option.

A $\frac{\pi}{3}, \frac{4\pi}{3}$

B $\frac{2\pi}{3}, \frac{5\pi}{3}$

C $\frac{\pi}{6}, \frac{7\pi}{6}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q067 MCQ

For $0 \leq \theta < 2\pi$, $\sin\theta = -1$ gives:

Choose the correct option.

A $\frac{\pi}{2}$

B $\frac{3\pi}{2}$

C π

D 0

Q068 MCQ

For $0 \le \theta < 2\pi$, $\cos\theta = 0$ gives:

Choose the correct option.

A $0, \pi$
B $\frac{\pi}{2}, \frac{3\pi}{2}$
C π
D $\frac{\pi}{4}, \frac{5\pi}{4}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q069 MCQ

The equation $\tan\theta = 0$ **has solutions in** $[0, 2\pi)$**:**

Choose the correct option.

A $0, \pi$
B $\frac{\pi}{2}, \frac{3\pi}{2}$
C $\frac{\pi}{4}, \frac{5\pi}{4}$
D $\pi, 2\pi$

Q070 **Written**

Solve $\cos\theta = \frac{\sqrt{3}}{2}$ **for** $0 \leq \theta < 2\pi$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q071 **Written**

Solve $\sin\theta = -\frac{\sqrt{3}}{2}$ **for** $0 \leq \theta < 2\pi$.

Q072 **Written**

Solve $\tan\theta = \frac{1}{\sqrt{3}}$ **for** $0 \le \theta < 2\pi$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q073 **Written**

Solve $2\cos\theta - 1 = 0$ **for** $0 \leq \theta < 2\pi$.

Q074 **Written**

Solve $2\sin\theta + \sqrt{2} = 0$ **for** $0 \leq \theta < 2\pi$**.**

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Quiz MCQ 1 Concept Quiz · MCQ

The radian measure of $240°$ is:

Choose the correct option.

A $\frac{2\pi}{3}$

B $\frac{4\pi}{3}$

C $\frac{3\pi}{4}$

D $\frac{5\pi}{3}$

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\sin\theta = -\frac{\sqrt{3}}{2}$, $0 \le \theta < 2\pi$, **then** θ **is:**

Choose the correct option.

A $\frac{\pi}{3}, \frac{2\pi}{3}$

B $\frac{4\pi}{3}, \frac{5\pi}{3}$

C $\frac{2\pi}{3}, \frac{4\pi}{3}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Quiz MCQ 3 **Concept Quiz · MCQ**

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Choose the correct option.

A 3π

B 6π

C 12π

D 18π

Quiz MCQ 4 **Concept Quiz · MCQ**

The general-angle family for $-300°$ is:

Choose the correct option.

A $60° + 360°\ n$

B $300° + 360°\ n$

C $-60° + 360°\ n$

D $120° + 360°\ n$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Quiz WRITTEN 5 **Concept Quiz · Written**

Solve $\cos\theta = -\frac{\sqrt{2}}{2}$ **for** $0 \le \theta < 2\pi$**.**

Quiz WRITTEN 6 **Concept Quiz · Written**

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Quiz WRITTEN 7 **Concept Quiz · Written**

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Quiz WRITTEN 8 **Concept Quiz · Written**

Evaluate $3\sin\frac{7\pi}{6} - 2\cos\frac{4\pi}{3}$.

Answer key

Use the question number shown on each page.

Q001 $(a)\ 435°\quad (b)\ -855°$

Q002 $300°\ :\ QIV$
$-450°\ :\ negative\ y-axis$

Q003 $60° + 360° \times n\quad ;\quad 150° + 360° \times n$

Q004 $(a)\ -30°\quad (b)\ 140°$

Q005 $500°\ :\ QI\qquad -120°\ :\ QIII$

Q006 $160° + 360° \times n\quad ;\quad 220° + 360° \times n$

Q007 30°; 405°; 110°; 310°; 300°; 240°

Q008 QII; QII; QIII; QI; QII; QII

Q009 $250° + 360°n;\ 200° + 360°n;\ 120° + 360°n;\ 290° + 360°n;\ 120° + 360°n;\ 340° + 360°n$

Q010 $120° + 360°n\quad ;\quad 260° + 360°n$

Q011 A

Q012 C

Q013 B

Q014 D

Q015 B

Q016 B

Q017 A

Q018 A

Q019 C

Q020 B

Q021 $\frac{7\pi}{2};\ -315°;\ L = 4\pi,\ S = 6\pi$

Q022 $\frac{8\pi}{5}, -\frac{20\pi}{9};\ 330°, -108°;\ L = \frac{9\pi}{2}, S = \frac{27\pi}{2}$

Q023 $\frac{3\pi}{5}, \frac{\pi}{5}, -\frac{4\pi}{3}, -\frac{5\pi}{4}, \frac{7\pi}{3}, \frac{20\pi}{9}, -\frac{19\pi}{6}, -15\pi$

Q024 210°, 144°, 120°, 240°, −225°, −90°, −990°, −7[illegible]

Q025 $(a)\ (7\pi, 21\pi);\ (b)\ (\frac{8\pi}{5}, \frac{16\pi}{5});\ (c)\ (\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 QI; QI; QII; QII; QIII; QIII; QIV; QIV

Q027 $h = 3\sqrt{5}\,\text{cm};\quad V = 36\pi\sqrt{5}\,\text{cm}^3$

Q028 A

Q029 B

Q030 B

Q031 B

Q032 B

Q033 C

Q034 A

Q035 D

Q036 D

Q037 $\frac{7\pi}{5}$

Q038 $L = 6\pi,\quad S = 24\pi$

Q039 $-\frac{\sqrt{3}}{2},\ -\frac{1}{2},\ \sqrt{3}$

Q040 $-\frac{\sqrt{2}}{2};\ \frac{1}{2};\ \frac{1}{\sqrt{3}};\ -\frac{\sqrt{3}}{2};\ -1;\ undefined$

Q041 $\frac{\sqrt{3}}{2}; -\frac{1}{2}; 0; -\frac{\sqrt{2}}{2}; \frac{1}{2}; -\frac{\sqrt{3}}{2}; -\sqrt{3}; 1; 0; -\frac{1}{2}, \frac{\sqrt{2}}{2}; \frac{\sqrt{3}}{2}; \frac{\sqrt{2}}{2}; -\frac{\sqrt{3}}{2}; -\frac{1}{2}; -\sqrt{3}; \frac{1}{\sqrt{3}}; undefined$

Q042 $\frac{3(1-\sqrt{2})}{2}\,\text{m}$

Q043 B

Q044 B

Q045 A

Q046 A

Q047 D

Q048 D

Q049 B

Q050 C

Q051 A

Q052 $\frac{1}{2}, -\frac{1}{2}, -1$

Q053 $-\frac{1}{2}, -\frac{\sqrt{3}}{2}, -\sqrt{3}$

Q054 $\frac{\sqrt{3}}{2}, -\frac{1}{2}, -\sqrt{3}$

Q055 -1

Q056 $\frac{\pi}{2}, \frac{3\pi}{2}$

Q057 $(a)\ \frac{7\pi}{6}, \frac{11\pi}{6};\ (b)\ \frac{\pi}{4}, \frac{7\pi}{4};\ (c)\ \frac{2\pi}{3}, \frac{5\pi}{3}$

Q058 $\frac{\pi}{4}, \frac{3\pi}{4};\ \frac{5\pi}{6}, \frac{7\pi}{6};\ \frac{\pi}{6}, \frac{7\pi}{6};\ 0, \pi;\ \frac{\pi}{3}, \frac{5\pi}{3};\ \frac{3\pi}{4}, \frac{7\pi}{4}$

Q059 $\frac{\pi}{6}, \frac{5\pi}{6};\ \frac{\pi}{3}, \frac{2\pi}{3};\ \frac{\pi}{2}, \frac{3\pi}{2};\ 0;\ 0, \pi;\ \frac{\pi}{3}, \frac{4\pi}{3};\ \frac{\pi}{2};\ \frac{5\pi}{4}, \frac{7\pi}{4};\ \frac{3\pi}{4}, \frac{5\pi}{4};\ \frac{2\pi}{3}, \frac{4\pi}{3};\ \frac{5\pi}{6}, \frac{11\pi}{6};\ \frac{\pi}{4}, \frac{5\pi}{4};\ \frac{4\pi}{3}, \frac{5\pi}{3};\ \frac{3\pi}{2};\ \pi;\ \frac{\pi}{6}, \frac{11\pi}{6}$

Q060 60°, 135°, 240°, 315°

Q061 A

Q062 B

Q063 A

Q064 A

Q065 A

Q066 B

Q067 B

Q068 B

Q069 A

Q070 $\frac{\pi}{6}, \frac{11\pi}{6}$

Q071 $\frac{4\pi}{3}, \frac{5\pi}{3}$

Q072 $\frac{\pi}{6}, \frac{7\pi}{6}$

Q073 $\frac{\pi}{3}, \frac{5\pi}{3}$

Q074 $\frac{5\pi}{4}, \frac{7\pi}{4}$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{3\pi}{4}, \frac{5\pi}{4}$

Quiz WRITTEN 6 $-390°;\ 330°$

Quiz WRITTEN 7 $L = 4\pi,\quad S = 20\pi$

Quiz WRITTEN 8 $-\frac{1}{2}$